



PROFILE

Driven by a passion for problem-solving, I actively seek opportunities to expand my skill set and leverage new tools to overcome challenges. My commitment to continuous learning has enabled me to contribute to diverse projects, where I focused on improving operational efficiency and creating practical, innovative solutions

CONTACT

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LANGUAGES

Working level Proficiency:

- English
- Marathi
- Hindi

Elementary level Proficiency:

- German

INTREST:

Computer Vision ,OS, Computer Networks, Computer Architecture,DL

SKILLS:

C/C++, Python, Bash,

Jayesh Khalane

B. Tech, Information Technology 2022-26
JSPM's Rajarshi Shahu College of Engineering, Pune
Affiliated to Savitribai Phule Pune University

WORK EXPERIENCE

• Research fellowship IIT Bombay [Jan 2026 – present]

Currently Working at Laboratory for Aeromechanical Autonomy. Project focus on two areas: cooperative / non cooperative satellite 6 DOF pose estimation and torque control for space manipulators.

Project Supervisor: Prof. Srikant Sukumar

• Research Intern -IIT Bombay [May 27th 2024 - July 19th 2024]

Worked on TurtleBot-based swarm project at ARMS Lab, IIT Bombay, Designed and implemented a novel communication-less multi-agent identification model enabling decentralized clustering without localization only using 2D LiDAR data as input.

Project Supervisor: Prof Leena Vachhani

PROJECTS

• ROBOCON Team, | Programming Member [Sep 2023 - Aug 2024]

Worked on autonomous robots involving LiDAR mapping, motion planning, localization, and navigation. Built pick-and-place mechanisms and integrated on board computers, and LiDAR sensors for real-time perception and control.

• Astro Tinker e-YRC | Team Lead [Sep 5th 2023 - Feb 27th 2024]

Worked on FPGA board to implement custom RISC-V 32-bit I-type processor, serial communication, PWM generation, pulse detection, UART, robust line following, pick up and place mechanism. Designed a complete robotic solution to the given problem statement by IIT-B robotic competition.

• Capstone Project | Team Lead [June 15th 2024 -June 2026]

Learning Based Methods for large scale 3D reconstruction from Satellite Imagery. Exploring deep learning and computer vision techniques for generating high-fidelity 3D models from satellite data like NeRF, MVS-NET, G-Splats and SV-Raster. Improved Existing COLMAP pipeline to reduce noise in reconstruction.

EDUCATION

Pratibha International School	SSC	2008-2020	77.20%
VVNP Jr college Pune	HSC	2020-2022	78.17%
JSPM'S RSCOE Pune	B Tech	2022-2026	8.84 CGPA

ACHIVEMENTS

- **Department Appreciation:** Awarded certificates for extra-curricular achievements during Second Year (SY) and Third Year (TY).
- **eYRC Semi-finalist** at the e-Yantra International Robotics Competition.Only team from the state to reach semi finals. Ranked 10-30 out of ~500 hundered teams